

Supplemental Material for “Improved Response Strategies for Unknown Behaviours in Multiagent Interaction”

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In this Supplementary Materials, we detail the frameworks of PSB and ACB, PSB operators, and shared functionalities. All implementations are included in our I-DID toolkit ¹.

A: PSB and ACB Implementation

We introduce PSB and ACB frameworks in Figs. 1 and 2, respectively, and elaborate on PSB operators and their shared functionalities with ACB.

A.1: [The framework of PSB and ACB]

Fig. 1 shows our PSB (Particle Swarm Optimization-enabled Behavior) framework. First, known policy trees become particles $\sigma = (p, v)$ forming pop (①). Fitness $F(\sigma)$ is calculated (②) to find local best $\bar{\sigma}$ and global best σ^* (③). Over N iterations, positions and velocities update (④-⑤), fitness recalculates (⑥), and best particles update (⑦). Particles move towards σ^* or $\bar{\sigma}$ (⑧). Finally, particles transform back to policy trees and decode into behaviors via *transform* (⑨), selecting top- K based on fitness (⑩). Behaviors, initialized with varied beliefs, optimize over iterations.

Fig. 2 presents our ACB (Ant Colony Optimization-enabled Behavior) framework. Ant populations and pheromone tables initialize (①-②). Pheromones update based on ant actions (③-④). Ants choose next actions epsilon-greedily, influenced by pheromones (⑤). After N iterations (⑥), the population converts to policy trees, selecting top- K behaviors (⑦-⑧).

A.2: [PSB Operators]

We introduce four PSB operators:

- *Minus* ($\ominus$): Computes particle velocity from positions.
- *Plus* ($\oplus$): Calculates new particle positions.
- *Times* ($\otimes$): Scales velocity v_1 by scalar ω .
- *Merging* ($\uplus$): Combines velocities v_1 and v_2 .

A.3: [The Common Operators in PSB and ACB]

Both PSB and ACB algorithms utilize four fundamental operators detailed in Alg. 2. The operators in PSB and ACB enable transformations between policy trees and particles/ants.

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¹<https://github.com/lamingic/SI-IDID>

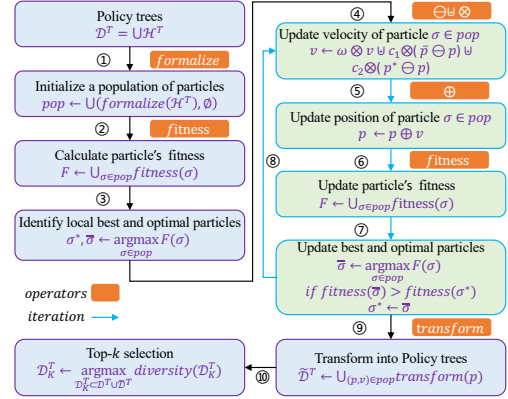


Fig. 1: PSO-enabled behavior generation framework comprises main components: initialization and updates of particles' positions and velocities.

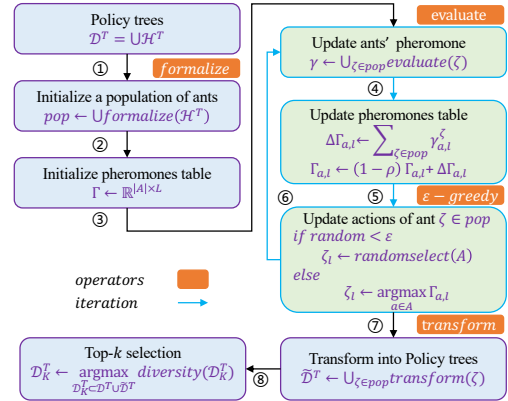


Fig. 2: The ACB algorithm generates new behaviors by updating the pheromone table, maintaining an ant population where each ant performs action sequences, updated upon selecting new actions.

- **Formalise:** Generates an action sequence from a policy tree.
- **Transform:** Converts action sequences to policy trees.
- **Fitness:** Evaluates particle sequences using GeNIe decision tool ².
- **Evaluate:** Calculates pheromone for ant policies.

²<https://www.bayesfusion.com>

Algorithm 1: The Four Operators in PSB

```

1 Function  $\ominus (p_1, p_2) \triangleleft v \leftarrow p_1 \ominus p_2$ :
2    $v \leftarrow \emptyset$ 
3   for  $l \in \{1, 2, \dots, |p_1|\}$  do
4     if  $p_1[l] \neq p_2[l]$  then
5        $v \leftarrow v \cup (l, p_1[l], 1/|A|)$ 
6     end
7   end
8 return  $v$ 
9 Function  $\oplus (p_1, v) \triangleleft p \leftarrow p_1 \oplus v$ :
10   $p \leftarrow p_1$ 
11  for  $(l, a, P) \in v$  do
12    if  $\text{random}() > P$  then
13       $a \leftarrow \text{randomselect}(A)$ 
14    end
15     $p[l] \leftarrow a$ 
16  end
17 return  $p$ 
18 Function  $\otimes (v_1, \omega) \triangleleft v \leftarrow \omega \otimes v_1$ :
19   $v \leftarrow \emptyset$ 
20  for  $(l, a, P) \in v_1$  do
21     $v \leftarrow v \cup (l, a, \min(\omega * P, 1))$ 
22  end
23 return  $v$ 
24 Function  $\uplus (v_1, v_2) \triangleleft v \leftarrow v_1 \uplus v_2$ :
25   $v \leftarrow \emptyset$ 
26  for  $(l, a, P) \in v_1$  do
27    for  $(l', a', P') \in v_2$  do
28      if  $l == l' \ \& \ a == a'$  then
29         $v \leftarrow v \cup (l', a', \min(P + P', 1))$ 
30      end
31      else
32         $v \leftarrow v \cup (l, a, P)$ 
33         $v \leftarrow v \cup (l', a', P')$ 
34      end
35    end
36  end
37 return  $v$ 

```

Algorithm 2: The operators in PSB and ACB

```

1 Function formalise ( $\mathcal{H}^T$ ):
2    $\triangleleft$  generate action sequence of tree  $\mathcal{H}^T$ 
3    $p \leftarrow \{a_l | a_l \leftarrow 0, \forall l \in \{1, 2, \dots, \frac{|\Omega|^T-1}{|\Omega|-1}\}\}$ 
4   for  $(h^T, i) \in \mathcal{H}^T$  do
5      $\{a_1, o_2, \dots, o_T, a_T\} \leftarrow h^T$ 
6     for  $t \in \{1, 2, \dots, T\}$  do
7        $j \leftarrow \frac{|\Omega|^{(t-1)}-1}{|\Omega|-1} + \lfloor \frac{i-1}{|\Omega|^{(t-1)}} \rfloor + 1$ 
8        $p[j] \leftarrow a_t$ 
9     end
10  end
11 return  $p$ 
12 Function transform ( $p$ ):
13    $\triangleleft$  generate policy tree from action
    sequence  $p$ 
14    $\mathcal{H}^T \leftarrow \bigcup h^T$ 
15   for  $(h^T, i) \in \mathcal{H}^T$  do
16      $\{a_1, o_2, \dots, o_T, a_T\} \leftarrow h^T$ 
17     for  $t \in \{1, 2, \dots, T\}$  do
18        $j \leftarrow \frac{|\Omega|^{(t-1)}-1}{|\Omega|-1} + \lfloor \frac{i-1}{|\Omega|^{(t-1)}} \rfloor + 1$ 
19        $a_t \leftarrow p[j]$ 
20       if  $t \neq T$  then
21          $k \leftarrow \lfloor \frac{(i-1)|\Omega|^t}{|\Omega|^{(T-1)}} \rfloor + 1$ 
22          $o_{t+1} \leftarrow o^k$ 
23       end
24     end
25      $h^T \leftarrow \{a_1, o_2, \dots, o_T, a_T\}$ 
26   end
27 return  $\mathcal{H}^T$ 
28 Function fitness ( $\sigma$ ):
29    $\triangleleft$  evaluate the fitness of action sequence  $\sigma$ 
30    $\mathcal{H}^T \leftarrow \text{transform}(\sigma)$ 
31   for  $s_i \in S$  do
32      $V_{\mathcal{H}^T}(s_i) \leftarrow R(s, a(\mathcal{H}^T)) +$ 
        $\lambda \sum_{s' \in S} Pr_{a(\mathcal{H}^T)}(s'|s) [\sum_{o \in O} Pr_{a(\mathcal{H}^T)}(o|s') V_{o(\mathcal{H}^T)}(s')]$ 
33   end
34    $\alpha_{\mathcal{H}^T} \leftarrow$ 
      $[V_{\mathcal{H}^T}(s_1), V_{\mathcal{H}^T}(s_2), \dots, V_{\mathcal{H}^T}(s_{|S|})]$ 
35    $f \leftarrow \sum_{s \in S} b^0(s) \alpha_{\mathcal{H}^T}(s)$ 
36 return  $f$ 
37 Function evaluate ( $\zeta$ ):
38    $\triangleleft$  evaluate the expected reward of action
    sequence  $\sigma$ 
39    $\mathcal{H}^T \leftarrow \text{transform}(\zeta)$ 
40    $ER \leftarrow \mathbb{R}^{|A| \times \frac{|\Omega|^T-1}{|\Omega|-1}}$ 
41   for  $(h^T, i) \in \mathcal{H}^T$  do
42      $\{a_1, o_2, \dots, o_T, a_T\} \leftarrow h^T$ 
43     for  $t \in \{1, 2, \dots, T\}$  do
44        $l \leftarrow \frac{|\Omega|^{(t-1)}-1}{|\Omega|-1} + \lfloor \frac{i-1}{|\Omega|^{(t-1)}} \rfloor$ 
45       for  $s \in S$  do
46          $b^t(s) \leftarrow$ 
            $\frac{Pr(o_l | s, a_l) \sum_{s' \in S} Pr(s | s', a_l) Pr(s' | b^{t-1})}{\sum_{s' s'' \in S} Pr(o_l | s'', a_l) Pr(s'' | s', a_l) Pr(s' | b^{t-1})}$ 
47       end
48        $ER(a_l, l) \leftarrow \sum_{s \in S} b^t(s) R(s, a_l)$ 
49     end
50   end
51 return  $ER$ 

```
